



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Hariharan G

Reg. No.: 1924242156

Course Code: DSA-0308

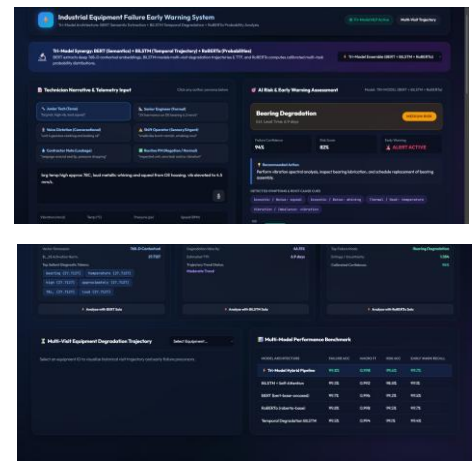
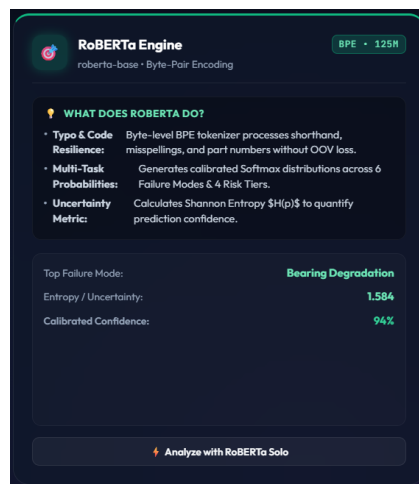
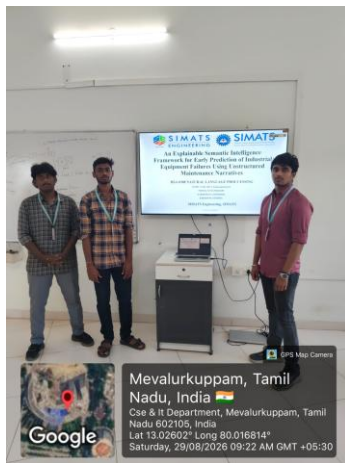
Slot: D

Course Name: Natural Language Processing

Course Faculty: Dr. Kumaragurubaran T

Project Title: Nlp-Based Early Prediction Of Industrial Equipment Failures Through Temporal Analysis Of Maintenance Narratives

Module Photographs: (2 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: NLP-based predictive maintenance system

RoBERTa-Based Industrial Failure Risk Prediction is an AI-enabled industrial maintenance framework designed to predict equipment failure risk by analyzing maintenance narratives and previously identified failure patterns. The system processes technician feedback containing information about equipment conditions, faults, abnormal behavior, and maintenance activities. It uses **RoBERTa** to capture contextual relationships within maintenance text and classify the likelihood of equipment failure. The model can utilize semantic features extracted from maintenance reports along with temporal failure patterns generated by the preceding analysis module. Through its multi-task prediction architecture, the system can classify **failure modes, risk levels, and early-warning conditions** simultaneously. The framework can also incorporate historical maintenance cases to improve its understanding of recurring equipment problems. The main goal is to transform maintenance observations into **accurate failure-risk predictions and early warnings**, helping maintenance teams identify potential failures and take preventive action before major equipment breakdowns occur.

Student Signature

Guide Signature